

CTBI Sanity Check

Independent verification of Dev's CTBI.csv (city-wide, 6 heatwave seasons, 91 days/year, 2021-2026). This is NOT a re-derivation of the formula — it is a behavioural check: does CTBI move the way physical reasoning says it should?

2021 season

- Start CTBI: 0.103 | Peak CTBI: 0.501 on 2021-06-14 | End CTBI: 0.342
- Max NCTL this season: 4.73 degree-hours
- Check 1 (CTBI rises from start to peak): 
- Check 2 (CTBI declines after peak, i.e. recovery): 
- Check 3 (near-zero NCTL in first 5 days of April): 
- **Result: PASS**

2022 season

- Start CTBI: 0.119 | Peak CTBI: 0.574 on 2022-06-25 | End CTBI: 0.418
- Max NCTL this season: 10.42 degree-hours
- Check 1 (CTBI rises from start to peak): 
- Check 2 (CTBI declines after peak, i.e. recovery): 
- Check 3 (near-zero NCTL in first 5 days of April): 
- **Result: PASS**

2023 season

- Start CTBI: 0.094 | Peak CTBI: 0.493 on 2023-05-21 | End CTBI: 0.304
- Max NCTL this season: 5.71 degree-hours
- Check 1 (CTBI rises from start to peak): 
- Check 2 (CTBI declines after peak, i.e. recovery): 
- Check 3 (near-zero NCTL in first 5 days of April): 
- **Result: PASS**

2024 season

- Start CTBI: 0.100 | Peak CTBI: 0.751 on 2024-05-25 | End CTBI: 0.349
- Max NCTL this season: 12.92 degree-hours
- Check 1 (CTBI rises from start to peak): 
- Check 2 (CTBI declines after peak, i.e. recovery): 
- Check 3 (near-zero NCTL in first 5 days of April): 
- **Result: PASS**

2025 season

- Start CTBI: 0.100 | Peak CTBI: 0.768 on 2025-05-23 | End CTBI: 0.252
- Max NCTL this season: 17.90 degree-hours
- Check 1 (CTBI rises from start to peak): 
- Check 2 (CTBI declines after peak, i.e. recovery): 
- Check 3 (near-zero NCTL in first 5 days of April): 
- **Result: PASS**

2026 season

- Start CTBI: 0.096 | Peak CTBI: 0.526 on 2026-06-05 | End CTBI: 0.443
- Max NCTL this season: 6.11 degree-hours
- Check 1 (CTBI rises from start to peak): 
- Check 2 (CTBI declines after peak, i.e. recovery): 
- Check 3 (near-zero NCTL in first 5 days of April): 
- **Result: PASS**

Cross-year check

- Correlation between a season's max NCTL and its peak CTBI: **0.944**
- Expected: positive and reasonably strong, since NCTL directly feeds CTBI via the $\gamma \cdot \text{NCTL}_{\text{norm}}$ term.
- **Result:  PASS**

Overall sanity verdict: ☒ ALL CHECKS PASS